Similarity & Proportional Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Rotations**

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(5)$.

Final Answer: 38. Substitute $x = 5$: $f(5) = 6(5) + 8 = 38$.

2 **[Foundational]** • **Target: Rotations**

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

3 **[Foundational]** • **Target: Dilations**

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

4 **[Foundational]** • **Target: Dilations**

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Final Answer: $x = 1$ and $x = 3$. Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

5 **[Foundational]** • **Target: Dilations**

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

6 **[Foundational]** • **Target: Dilations**

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(3)$.

Final Answer: 15. Substitute $x = 3$: $f(3) = 2(3) + 9 = 15$.

7 **[Foundational]** • **Target: Dilations**

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Dilations

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(4)$.

Final Answer: 17. Substitute $x = 4$: $f(4) = 2(4) + 9 = 17$.

9 [Intermediate] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

10 [Intermediate] • Target: Dilations

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(5)$.

Final Answer: 19. Substitute $x = 5$: $f(5) = 2(5) + 9 = 19$.

11 [Intermediate] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

12 [Intermediate] • Target: Transformations, congruence, and similarity

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 45^\circ) = 84^\circ$.

13 [Intermediate] • Target: Transformations, congruence, and similarity

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 10 cm. Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

14 [Intermediate] • Target: Transformations, congruence, and similarity

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 6. Substituting $x = 1$ into $y = 3x + 3$ gives $y = 3(1) + 3 = 6$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 77°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 52^\circ) = 77^\circ$.

16 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 15 cm. Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

17 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 10.01 Coordinate Geometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 9. Substituting $x = 2$ into $y = 3x + 3$ gives $y = 3(2) + 3 = 9$.

18 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 11.01 Probability

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 70°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 59^\circ) = 70^\circ$.

19 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 2.01 Lines & Angles

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

20 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 3$ into $y = 3x + 3$ gives $y = 3(3) + 3 = 12$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(2)$.

a. 14 ✓

b. 18

c. 8

d. 18

Substitute $x = 2$: $f(2) = 4(2) + 6 = 14$.

22 [Foundational] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

23 [Foundational] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(5)$.

a. 26 ✓

b. 30

c. 20

d. 30

Substitute $x = 5$: $f(5) = 4(5) + 6 = 26$.

24 [Foundational] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(2)$.

a. 17 ✓

b. 22

c. 10

d. 21

Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

25 [Foundational] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

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26 [Foundational] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(5)$.

a. 32 ✓

b. 37

c. 25

d. 36

Substitute $x = 5$: $f(5) = 5(5) + 7 = 32$.

27 [Foundational] • Target: Rotations

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(2)$.

a. 20 ✓

b. 26

c. 12

d. 24

Substitute $x = 2$: $f(2) = 6(2) + 8 = 20$.

28 [Foundational] • Target: Rotations

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

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30 [Intermediate] • Target: Translations

A data set recorded for translations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

31 [Intermediate] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(4)$.

a. 22 ✓

b. 26

c. 16

d. 26

Substitute $x = 4$: $f(4) = 4(4) + 6 = 22$.

32 [Intermediate] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

33 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

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34 [Intermediate] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(3)$.

a. 22 ✓

b. 27

c. 15

d. 26

Substitute $x = 3$: $f(3) = 5(3) + 7 = 22$.

35 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

36 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

37 [Intermediate] • Target: Rotations

A data set recorded for rotations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Rotations • Cross-Lesson: 11.01 Probability

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(4)$.

a. 32 ✓

b. 38

c. 24

d. 36

Substitute $x = 4$: $f(4) = 6(4) + 8 = 32$.

39 [Challenging] • Target: Dilations • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

40 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

41 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77° ✓

b. 87°

c. 69°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 52^\circ) = 77^\circ$.

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42 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70° ✓

b. 80°

c. 62°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 59^\circ) = 70^\circ$.

43 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

44 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15 ✓

d. 18

Substituting $x = 4$ into $y = 3x + 3$ gives $y = 3(4) + 3 = 15$.

45 [Challenging] • Target: Compositions of transformations • Cross-Lesson: 9.01 Transformations & Symmetry

For the model $f(x) = 4x + 4$ in compositions of transformations, evaluate $f(3)$.

a. 16 ✓

b. 20

c. 12

d. 20

Substitute $x = 3$: $f(3) = 4(3) + 4 = 16$.

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